A computer-assisted dimension bound for compact arithmetic hyperbolic Coxeter polytopes

Coxeter Polytopes Frontier Project

Proof translation, 18 October 2026

Abstract

We give an exposition of the completed internal proof that a compact arithmetic hyperbolic Coxeter polytope has dimension at most 21. The proof combines published high-dimensional field restrictions with complete local arithmetic reductions and exact rational angle weights. The difficult golden-field step excludes dimension 22 by a vertex bound $21977/500 < 44$; the strict lower bound 44 comes from the h-vector of a simple compact polytope. We explain why the local classification covers arbitrary supporting walls, why singular intermediate states must remain payment obligations, and why the normalized cross-component weights admit exact forest maximization. This document translates the existing proof and its certificate interfaces; it does not report a new geometric or arithmetic census. The result is internally checked, not externally refereed or fully formalized in a proof assistant. No compact dimension 21 example or compact-only sharpness is asserted.

1 Statement, scope, and proof architecture

A Coxeter polytope means a finite-sided convex hyperbolic polytope whose facet dihedral angles are integral submultiples of π. Arithmeticity refers to the lattice generated by its facet reflections and its actual Vinberg ground field, not to a field merely containing displayed entries.

Theorem 1.1. *Every compact arithmetic hyperbolic Coxeter polytope has dimension at most 21.*

The internal proof of this theorem is complete: the bound-21 composition, its independent scope checker, and the geometric audit identify no unresolved mathematical step. Historical immutable reports use "proof candidate" to record the absence of external validation; this is not an unfilled step in the internally completed argument. No priority claim is made here.

The proof has three kinds of inputs, which must not be confused.

1. Published geometry and arithmetic: compact simplicity and face correspondence, finite, affine and Lannér classifications, arithmetic cyclic-product conditions, the compact dimension ceiling 29, the high-dimensional ground-field list, and the nonnegative symmetric h-vector theorem.

2. Internal mathematical arguments: projection to actual faces, complete-star rank, first-touch reduction, all-size attachment closure, whole-component weight normalization, and the strict counting contradiction.

3. Finite exact certificates: every member of a proved finite domain is checked for its necessary arithmetic conditions, literal reduction maps, face payments, and finite-forest upper bounds. Floating optimization proposes weights but is not used to certify them.

The detailed assumption/reproduction map is `docs/compact_arithmetic21_proof_map.md`. The final composition is `scripts/fresh_arithmetic_dimension21_composition_20261016.py`; the independent final scope check is `scripts/fresh_arithmetic_dimension21_scope_check_20261016.py`. Neither small program by itself establishes the geometric premises below.

2 The strict weighted-face contradiction

Let P be a simple compact n-polytope, and write f_i for its numbers of i-faces. A planar angle is a pair (v, F) where v is a vertex of a 2-face F. At v it corresponds to an unordered pair of the n incident facets, equivalently a marked pair in its finite Coxeter diagram.

Lemma 2.1. *Suppose nonnegative angle weights satisfy*

$$\sum_{v \in F} \mathrm{wt}(v, F) \geq 5 - k \quad \textit{for every } k\textit{-gonal 2-face } F.$$

Then the average total vertex weight is strictly greater than

$$\beta(n) = \begin{cases} n(n-6)/8, & n \textit{ even}, \\ (n-1)(n-5)/8, & n \textit{ odd}. \end{cases}$$

In particular, a uniform vertex upper bound at most $\beta(n)$ is impossible.

Proof. Double counting angles gives

$$\sum_{v} \mathrm{wt}(v) \geq 5f_2 - \binom{n}{2} f_0.$$

For the nonnegative symmetric h-vector of the simplicial dual,

$$f_0 = \sum_{i=0}^{n} h_i, \qquad f_2 = \sum_{i=0}^{n} \binom{n-i}{2} h_i.$$

Pair i and $n - i$. The minimum paired coefficient is $n(n-2)/8$ for even n and $(n-1)^2/8$ for odd n. The endpoint coefficients are strictly larger, and $h_0 = h_n = 1$. Consequently f_2/f_0 is strictly larger than that minimum. Substitution proves the claimed formula. Strictness does not require every middle h_i to be positive. □

It suffices to pay 2 on every triangle and 1 on every quadrilateral; nonnegativity handles every $k \geq 5$. We never assume a face is itself a Coxeter polytope: its angles are evaluated from the ambient finite vertex forest.

3 Actual faces and complete supporting stars

Use the Lorentz convention $(n, 1)$, inward chamber inequalities $\langle x, r_i \rangle \leq 0$, and spacelike outward roots with pairwise nonpositive products. A finite corner has positive-definite Gram matrix.

Lemma 3.1. *An elliptic principal root set determines an actual face of the compact chamber. A positive affine principal set is incompatible with compactness.*

Proof. For the elliptic set let B be its positive-definite Gram matrix and h a strict interior timelike vector. Since B is a positive-definite Z-matrix, B^{-1} is entrywise nonnegative. Put $c = B^{-1}(\langle h, r_i\rangle)_i \le 0$ and $h_0 = h - \sum_i c_i r_i$. It is orthogonal to the chosen roots,

$$h_0^2 = h^2 - (\langle h, r_i\rangle)^T B^{-1}(\langle h, r_i\rangle) < 0,$$

and has strictly negative product with every other chamber root. This supplies a relative-interior face point. For an affine connected principal set, its positive kernel produces a nonzero null vector: it cannot be the zero vector because pairing the positive relation with h is negative. Its products with all other roots are nonpositive. It gives an ideal point of the chamber, contrary to compactness. □

For a quadrilateral 2-face, let W be its full $n-2$ supporting roots and let b_0, b_1, b_2, b_3 be its cyclic boundary roots. Every $W + \{b_i, b_{i+1}\}$ is spherical. Each opposite extension $W + \{b_0, b_2\}$ or $W + \{b_1, b_3\}$ is indefinite by Lemma 3.1; an elliptic opposite extension would make opposite facets meet on this face. Select a minimal indefinite subdiagram in each. It must contain both relevant black roots because deleting either leaves a principal subset of a finite corner. These are arithmetic Lannér pieces; the order bound 5 is a published input.

The union of the two pieces is only an initial core. Shared white roots, all black/white identifications, and every cross contact are retained. Every unspecified pair belongs to some finite corner and therefore has a finite label. Opposite dotted contacts remain arithmetic parameters.

Lemma 3.2. *A k-root core with a positive-definite corner of order $k-2$ and a negative direction has nullity at most 1. The complete $n+2$-root quadrilateral star has nullity exactly 1.*

Proof. The positive index is at least $k-2$, and the physical ambient form has negative index 1. This proves the first assertion. A complete star has $k = n+2$ and lies in Lorentz dimension $n+1$, so its Gram is singular; the first assertion makes its nullity exactly 1. □

If a core is singular and has a negative direction, its Gram kernel is an actual relation among its normals: a Lorentz subspace containing a timelike vector is nondegenerate. Every later contact row must annihilate that kernel. A nonnegative root relation is impossible by the strict interior point. These are exact geometric rejection tests, not numerical rank heuristics.

Every additional white component touching a core has a first touching root, followed by an ordering along its connected white edges. Enumerating all such one-white contacts is therefore enough for a complete attachment queue. If no unpaid contact survives, all remaining roots are orthogonal spectators. Marked connected components in the core are then entire in the full finite corner. A nonsingular core plus only positive-definite orthogonal spectators cannot acquire the nullity required by Lemma 3.2.

A singular state may stop with only orthogonal spectators *even if it also has possible touching extensions*. Thus every singular intermediate state must be retained as a separate payment obligation. Keeping only terminal sealed states would be an incorrect completeness argument.

4 The two fields with an all-rank bound 20

The arithmetic condition used throughout is integral cyclic products in the actual ground field and positive-semidefinite conjugate Grams at every nonphysical embedding. Finite contacts and Lannér pieces are classified using these conditions, rather than a chosen label cutoff.

4.1 The field $\mathbb{Q}(\sqrt{2})$

The complete finite alphabet is $2, 3, 4, 6, 8$: the two-cycle $2 + 2\cos(2\pi/m)$ must belong to the field. The Lannér list consists of an arbitrary permitted dotted pair, nine triangles, and the two chordless cycles C_4, C_5 with one 4-edge and all remaining edges 3. The triangle labels are

$$(2,3,8),(2,4,8),(2,6,8),(2,8,8),(3,3,4),$$
$$(3,8,8),(4,4,4),(4,6,6),(4,8,8).$$

The field catalogue proves completeness through cyclotomic degree, squareclass parity and exact conjugate determinants.

Assign a base unit to every marked pair at graph distance at most 2. For the unpaid case, all marked minimal unions are exhausted: 51 marked pieces, 6,010 shared-white identifications, 115 literal unions and 52 marked classes. Forty classes have two negative directions and one fails cyclic field membership. Nine finite-label cores and two dotted families remain. Complete one-white domains leave seven singular finite cores sealed; a further nonsingular core can only extend to a sealed core. The last finite core admits only short-arm path extensions of D_5 to E_6, E_7, E_8, or a first affine obstruction. If its two black endpoints are not connected through new whites, elimination gives reduced unit squares

$$p, q \in \{1, 3/4, 2/3, 5/8\},$$

and complete-star singularity requires

$$(p + q - 4pq) + (1 - p - q)\sqrt{2} = 0.$$

This would force $p + q = 1$, contradicting $p + q \geq 5/4$. Connecting paths give literal sealed-core maps, not additional unbounded families.

For the dotted cores, every outside contact is at one of the two dotted endpoints. Eliminating a white component joining them leaves $0 < p, q \leq 3/4$ and increases the effective norm-two dotted magnitude to $r_{\text{eff}} > 2$. In the single-contact case the rank identity gives

$$r_{\text{eff}}^2 = q(4p - 1 + \sqrt{2}) \leq \tfrac{3}{4}(2 + \sqrt{2}) < 4.$$

In the double-contact case it gives

$$r_{\text{eff}} \leq \tfrac{1}{2} + \sqrt{2} < 2.$$

Both are contradictions. The independent pendant classification covers all remaining disconnected tails. Thus no arbitrary white component has been silently replaced by a bounded path search.

Add whole-component bonuses to the base rule:

Component	Marked pairs	Total bonus
F_4	two path endpoints	1
D_5	long tip with either short tip	1
E_6	near/far nodes on the two equal arms	1
E_7, E_8	specified unequal-arm near/far pair	1/4 each

The named unequal-arm pair is the near node of the longest arm and the length-two-arm tip. The D_5, E_6 pairs each receive halves. The sealed-corner tables give every exceptional quadrilateral at least 1. Triangles receive at least 2: a three-black minimal Lannér triangle has two nonright edges, while a four- or five-cycle has at least two black cyclic gaps of length at most 2. These paths remain in the corresponding finite corners under additional white extensions.

The distance-two count on a rank-r path is $2r-3$, and on a D/E tree it is $2r-2$. The only deficits below 2 from $2r$ after bonuses are $1, 1, 7/4, 7/4$ on D_5, E_6, E_7, E_8. For total rank $n \geq 9$, a single component has deficit at least 2, or at least two components have combined deficit at least 2. Every vertex therefore has weight at most $2n-2$.

4.2 The cubic field $K_7 = \mathbb{Q}(\cos(2\pi/7))$

Here the arithmetic Lannér classification leaves orders 2 and 3. A triangle has a distinguished 7- or 14-edge; without such an edge its entries and determinant field cannot generate K_7. A high edge in a finite diagram is isolated. Complete minimal-union finite-corner and short-path conditions leave nine supports. The arbitrary-white argument projects off the two opposite positive black blocks: white neighbors have negative residual squares; a 14-edge or a dense contact would give two negative directions. All contacts between the two surviving white neighbor sets are nonright, so the white forest forces one set to be a singleton and the other to have size 1 or 2. Off-core components attach only once and have rational Schur diagonal corrections.

In the six-node case, full-star singularity would give a nonzero rational quadratic in the degree-three number c^2, which is impossible. In the seven-node case write the remaining white matrix as

$$H = \begin{pmatrix} p & -\delta & -\varepsilon \\ -\delta & q & 0 \\ -\varepsilon & 0 & r \end{pmatrix} > 0, \qquad L = \delta\varepsilon\beta\gamma.$$

Cycle-field membership puts L in $K_7 \cap \mathbb{Q}(\sqrt{2}) = \mathbb{Q}$. The coefficient of the linear term in the reduced cubic determinant is $(\det H - 8L)/8$. Consequently

$$8L = \det H \leq 1 - \delta^2 - \varepsilon^2 \leq \tfrac{1}{2} \leq 8L.$$

Equality removes every external branch and forces the displayed contacts to be simple. The other coefficient forces the remaining contact simple. The two exceptional paying corners are entire D_5 factors.

Use the distance-two base, one additional unit on isolated $I_2(7), I_2(14)$, and halves on the two long-tip/short-tip pairs of whole D_5. Triangles receive 2 at a high-edge corner. Quadrilaterals receive 1 from a short corner or the two D_5 halves. The same component-deficit argument gives vertex weight at most $2n-2$ for $n \geq 6$.

Proposition 4.1. *For either $K = \mathbb{Q}(\sqrt{2})$ or $K = K_7$, a compact arithmetic Coxeter polytope has dimension at most 20.*

Proof. The finite certificates and all-size arguments just described establish the hypotheses of Lemma 2.1. At $n = 21$ its strict lower bound is 40, equal to the upper $2n-2$. More generally, for $n = 21 + 2u$ or $n = 22 + 2u$, subtraction of $2n-2$ from $\beta(n)$ gives respectively $u^2/2 + 5u$ and $u^2/2 + 11u/2 + 2$. Both are nonnegative for every integer $u \geq 0$. Strictness excludes every $n \geq 21$. □

The finite numerical identities in this section are certified by the independent square-root-field quadrilateral and weight reviewers and the cubic-seven global reviewer listed in the proof map. Their all-size geometric proofs are essential; the tables alone would not suffice.

5 Complete golden-field reduction

Put $\phi = (1+\sqrt{5})/2$ and $K = \mathbb{Q}(\phi)$. The finite alphabet is $2, 3, 4, 5, 6, 10$. The exact arithmetic Lannér catalogue has 12 triangles, six order-four types, four order-five types, and the dotted parameter

$$t \in \mathbb{Z}[\phi], \qquad t > 4, \quad 0 < t' < 4.$$

The prime denotes the other real embedding. The fixed baseline is

$$b(C; i, j) = \mathbf{1}_{d_C(i,j) \le 2} + \mathbf{1}_{C \cong H_4}\, \mathbf{1}_{\{d_C(i,j)=3\} \text{ or } \{ij \text{ is the 5-edge}\}}.$$

Here C is an *entire* connected component. Short paths persist in finite extensions, and H_4 has no larger connected spherical extension. The 76 minimal triangle markings give baseline at least 2.

For an unpaid quadrilateral, the no-distance-two minimal-union reduction has 5,440 literal supports and 1,180 marked classes. Arithmetic and signature filters leave 362 finite-label cores and 34 dotted families. After persistent payments and affine exclusions, the obligations are:

Reduction	Number
Original sealed singular rows	112
Singular first attachments	147
Marked singular frontier rows	51
Further closure rows	17
Singular dotted-queue stopping rows	264
Singular finite-queue stopping rows	2,070
Total zero-touch obligations	2,661

Every listed singular row is retained even if it also admits touching extensions. Nonsingular orthogonally sealed rows are rank-excluded.

The finite queue has 5,172 states and 51,599 complete one-white profiles. The dotted queue has 438 states, including 423 below its geometric boundary, and 5,146 next-white profiles. Each retained child has a literal full-label bijection; marked black pairs and exact dotted parameters are preserved. The maximum 31 nodes is justified by $n + 2 \le 31$ after the published compact ceiling 29, not by a resource cutoff.

For the 24 residual dotted families, deleting the two dotted endpoints leaves a nonsingular Lorentz block B. The endpoint Schur complement must be positive semidefinite in any ambient Lorentz extension. Writing it as $\left(\begin{smallmatrix} a & -x-c \\ -x-c & b \end{smallmatrix}\right)$ gives $x^2 \le 2ab + 2c^2$. Arithmeticity bounds the corresponding integral two-cycle $t = A + B\phi$ simultaneously at the two embeddings. This gives a finite integer box, containing 532 tested candidates and 32 admissible parameter seeds. Hence the dotted queue is exact and finite; no distance sampling replaces a continuous parameter argument.

The first-touch argument of Section 3 proves that every actual unpaid quadrilateral reaches a retained stopping row or a persistent baseline payment. Remaining orthogonal white components are arbitrary spectators. This is the geometric content of complete coverage, beyond a statement that the stored queue has no pending jobs.

6 Exact whole-component weights at rank 22

Let a, b, c count entire $A_1, I_2(5), H_3$ components of a finite vertex forest. The rational coefficient table is specified by `results/fresh_sqrt5_dimension22_h3_cross_20261016_cap2.json`:

its `variable_order` indexes `certificate.exact_primal`. The terminal coordinate records the vertex upper; the preceding 30,169 coordinates are nonnegative rational bonuses. There are 232 connected marked-pair orbits,178 rooted A_1 destinations,97 rooted $I_2(5)$ destinations and six rooted H_3 destinations. Orbit graphs and marks, not opaque string keys alone, define the table.

The coefficient context is $(a, \min(b, 2), \min(c, 1))$. Add a connected bonus to a baseline-zero marked pair in the same entire component. For pairs in distinct components use the following disjoint rules:

Source	Excluded destination types	Payment denominator
A_1	A_1	a
$I_2(5)$	$A_1, I_2(5)$	$2b$
H_3	$A_1, I_2(5), H_3$	$3c$

The numerator is the rooted destination coefficient in that context. All omitted coefficients are zero. In particular an A_1–H_3 pair belongs only to the first rule, and an $I_2(5)$–H_3 pair only to the second. Denominators use *actual* counts, not capped counts.

Lemma 6.1. *At fixed exact (a, b, c) the total vertex charge is additive over entire finite components.*

Proof. For any eligible destination root there are respectively $a, 2b, 3c$ source roots. Their contributions sum to the numerator once. Disjointness prevents double counting. Connected bonuses and the baseline are already componentwise. A hub component may still receive incoming payments from an earlier hub family; these are included in its own additive cost. □

Let $C_T(a, b, c)$ denote the resulting exact cost of type T. Fix the three hub counts and put $r = 22 - a - 2b - 3c$. The residual forest contains none of the three hub types. With $D(0) = 0$ and unattainable ranks assigned $-\infty$, use

$$D(s) = \max_{T \notin \{A_1, I_2(5), H_3\},\, \mathrm{rk}\, T \le s} \{D(s - \mathrm{rk}\, T) + C_T(a, b, c)\}.$$

This recurrence includes every multiset of finite components by removing one final component; conversely every transition appends a permitted component. Add the costs of the fixed hubs at the end. Rank 1 is the sole unattainable residual rank, since A_2, A_3 realize every integer at least 2.

For a k-node stopping core at ambient rank 22, spectator rank is $24 - k$. Enumerate every common external triple (e, f, g) with

$$e + 2f + 3g \le 24 - k, \qquad 24 - k - e - 2f - 3g \ne 1.$$

The same external triple is added to each corner's own internal counts before the coefficient context and denominator are evaluated. Choosing spectators independently at the four corners would not prove a face inequality.

Finite certificate lemma 6.2. *The specified rank 22 rational rule has the following exact properties.*

1. *All 76 triangle baseline totals are at least 2.*
2. *All 2,661 stopping identities are represented. Precisely 1,634 fit rank 22; their 91,726 common spectator triples have nonnegative corner weights and quadrilateral total at least*

$$\frac{560003}{560000} > 1.$$

3. *All 71 finite component types and 410 exact hub-count contexts have vertex total at most* 21977/500.

The independent reviewer reconstructs finite graphs and rooted/marked orbits, tests every literal label-preserving map, and runs a type-first unbounded-coin recurrence in reverse component order, distinct from the producer's rank-first recurrence. The complete coverage program reevaluates all older stopping branches under this new rule, rather than assuming an old numerical payment transfers. The certificate lemma is a finite exact computation on an explicitly complete domain, with the published classifications and Section 3 supplying its geometric meaning.

At $n = 22$, Lemma 2.1 gives strict average weight greater than 44. Certificate 6.2 gives at most $21977/500 = 43.954$. This contradiction excludes dimension 22. The already completed dimension-specific rules exclude all larger remaining dimensions:

n	Exact vertex upper	Strict lower threshold $\beta(n)$
22	21977/500	44
23	485211/10000	99/2
24	267527/5000	54
25	59	60
26	31017/500	65
27	67009/1000	143/2
28	230/3	77
29	84	84

Each row has its own fixed rational rule and complete current-rule face payments on the same exhaustive arithmetic reduction. The equality at 29 is contradicted by the strict lower bound. One does not extrapolate the rank 22 coefficients to higher ranks.

Proposition 6.3. *Compact arithmetic Coxeter polytopes over $\mathbb{Q}(\sqrt{5})$ have dimension at most 21.*

Proof. The table excludes 22 through 29. The published compact ceiling excludes higher dimensions. □

7 Field composition and the noncompact corollary

Proof of Theorem 1.1. Suppose $n \geq 22$. Vinberg's compact ceiling gives $n \leq 29$. The published high-dimensional field restriction [1, 2] leaves $\mathbb{Q}, \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{5}), K_7$. The rational field is impossible in this compact range: arithmetic reflection lattices are of simplest type, and a rational indefinite form in at least five variables is isotropic by Meyer, so its arithmetic quotient is noncompact. Proposition 4.1 excludes the two bound 20 fields, and Proposition 6.3 excludes the golden field. This covers all $8 \cdot 3 = 24$ compact field/dimension branches, using the field restriction only in its stated range $n \geq 22$. □

Combining this with Esselmann's published noncompact arithmetic bound [4] $n \leq 21$ gives the same bound for all finite-volume arithmetic reflection groups. Borcherds' noncompact dimension 21 example [3] makes that combined bound sharp. It does not make the compact-only bound sharp. No result here concerns arbitrary nonarithmetic compact chambers beyond the separately known ceiling 29.

8 Verification boundaries and further work

The complete proof is computer-assisted, not a mere manifest comparison. Its mathematical parts are the reductions and counting implications above; its finite parts are exact field/inertia, graph-domain, map, weight and forest checks. Hashes identify which finite objects were checked. They do not prove geometric completeness. A separate final scope checker matches all 2,661 obligation identities,91,726 common triples,410 contexts and 24 field/dimension branches to the originating catalogues.

This translation does not rerun the large classification queues. The reproduction map separates a quick immutable binding/scope replay from full independent finite-check replays. Published inputs may be axioms in a future proof-assistant layer. Internal certificate lemmas must instead be proved, reflected into checked computation, or explicitly reported as remaining formalization assumptions. A Lean theorem importing the entire field bounds as axioms would certify only the final composition, not this whole computer-assisted proof.

Compact dimension 21 remains an existence/nonexistence problem; later field-specific work reduces its possible ground field to $\mathbb{Q}(\sqrt{5})$. The current rank 13 incidence model has a verified dual lower bound above its target 12, so additional valid incidence information is required; reoptimizing the same matrix is insufficient. In the general compact lane, excluding dimension 29 via this weighting framework requires a vertex upper at most $\beta(29) = 84$ instead of the classical $3n = 87$. A concrete next step is to classify the maximizing rank 29 forests under the current geometric rule and seek whole-component exchanges whose face proofs preserve every triangle and quadrilateral payment. These are research directions, not improvements asserted by this manuscript.

References

[1] E. B. Vinberg, *Absence of crystallographic groups of reflections in Lobachevskii spaces of large dimension*, Transactions of the Moscow Mathematical Society 47 (1984),75–112; original Russian text, `https://www.mathnet.ru/php/getFT.phtml?jrnid=mmo&option_lang=rus&paperid=444&what=fullt`.

[2] V. V. Nikulin, *On ground fields of arithmetic hyperbolic reflection groups*, especially Introduction and Sections 2,3, `https://arxiv.org/abs/0708.3991`.

[3] M. Belolipetsky, *Arithmetic hyperbolic reflection groups*, especially Lemma 2.1 and Sections 1,7, `https://arxiv.org/abs/1506.03111`.

[4] F. Esselmann, *Über die maximale Dimension von Lorentz-Gittern mit coendlicher Spiegelungsgruppe*, Journal of Number Theory61 (1996),103–144, `https://doi.org/10.1006/jnth.1996.0141`.

[5] Exact internal sources and independent reviews, fixed 16 October 2026: the path-indexed inventory in `docs/compact_arithmetic21_proof_map.md`. Historical status wording is preserved in the frozen files; this translation supplies no new priority, external-referee, or complete proof-assistant claim.